Extensive-form games

Chomp: $m \neq n$

Theorem

For every finite $n \times m$ board (with $n > 1$ or $m > 1$), Player I, who has the opening move, has a winning strategy.

Chomp: $m \neq n$

- The game satisfies Theorem 1, and it cannot end in a draw.
- Hence, either of the two possibilities holds:
  1. Player I has a winning strategy.
  2. Player II has a winning strategy.
- We will prove the following claim:

Claim

For every finite $n \times m$ board (with $n > 1$ or $m > 1$), if Player II has a winning strategy, then Player I also has a winning strategy.

Chomp: $m \neq n$

- This claim implies that Player II cannot have a winning strategy. Hence, by Theorem 1, Player 1 must have a winning strategy.

Games with chance moves

- Sometimes we come across situations in which transitions from state to state depend on chance factors.
- You may think of a toss before the start of a game or the transition from one state to another could depend on some factors such as rain or earthquake.
- To accommodate this feature, our model is expanded by labeling some of the vertices in the game tree (V, E, x_0) as chance moves.
- The edges emanating from vertices corresponding to chance moves represent the possible outcomes of a lottery.

Definition

Definition

A game in extensive form (with perfect information and chance moves) is a tuple

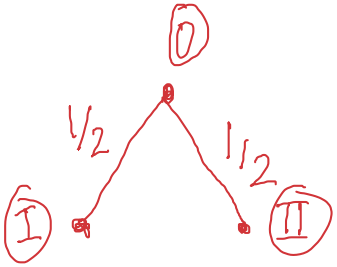
$$\Gamma = \langle N, V, E, x_0, (V_i)_{i \in N \cup \{0\}}, (p_x)_{x \in V_0}, (u_i)_{i \in N} \rangle$$

where

- N is a finite set of players.
- (V, E, x_0) is a directed rooted tree, called the game tree.
- $(V_i)_{i \in N \cup \{0\}}$ is a partition of the set of vertices that are not leaves among the agents and nature (Player 0), i.e, $V \setminus Z$.
- For every vertex $x \in V_0$, p_x is a probability distribution over the edges outgoing from x.
- $(u_i)_{i \in N}$ is the collection of utility functions where $u_i : Z \to \mathbb{R}$ for all $i \in N$.

Example

- Consider the following game:



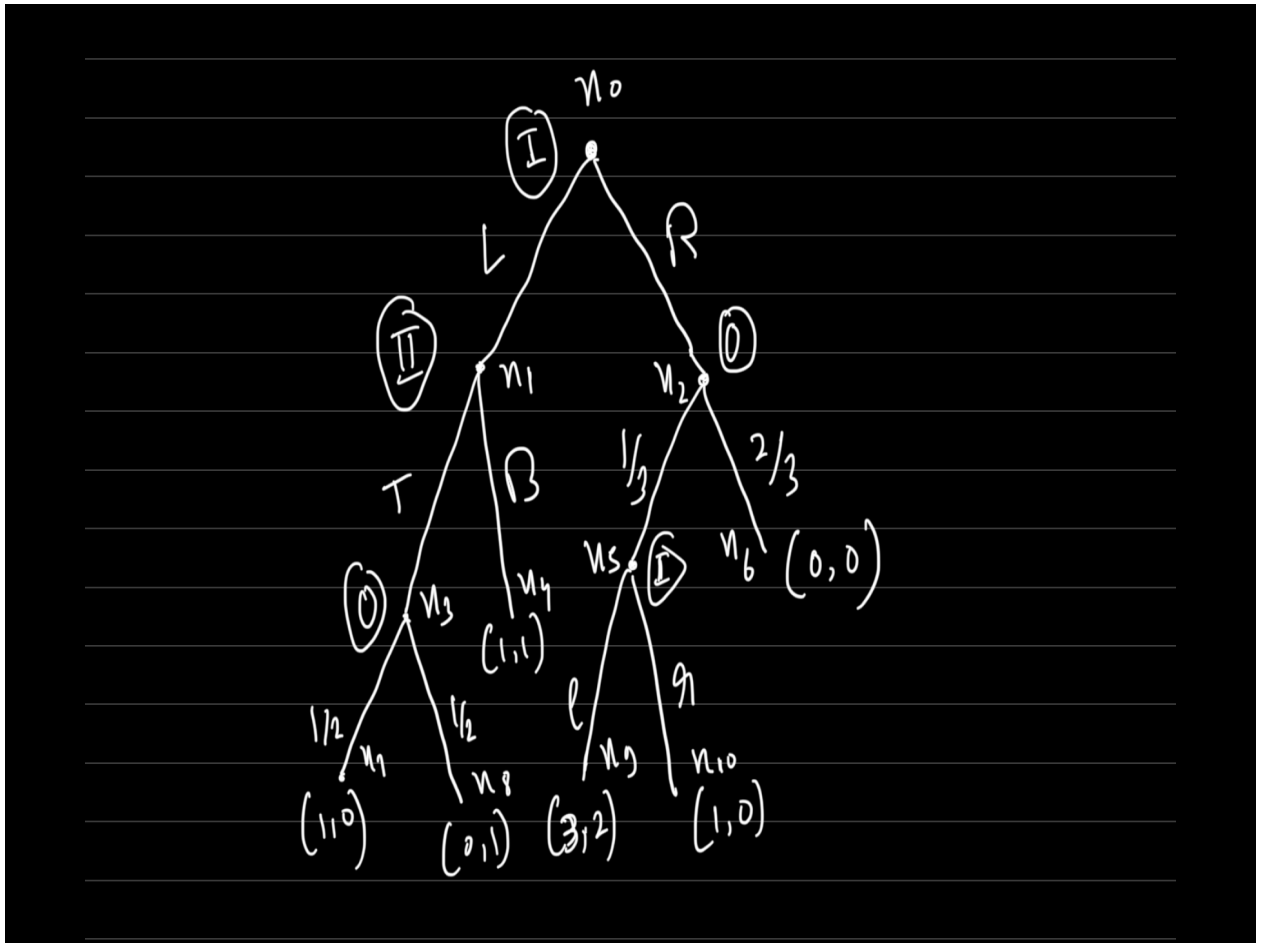


Figure: Figure 1

Chance nodes

- We make the assumption that the probabilities at the chance nodes are known to the players, and there is no player who still has to move after the chance move and is not informed about the outcome of the chance move.

Play of the game

- Games with chance moves are played similarly to games without chance moves.
- Only difference is that at vertices with chance moves a lottery is implemented, to determine the action to be undertaken at that vertex.

Utilities at a strategy profile

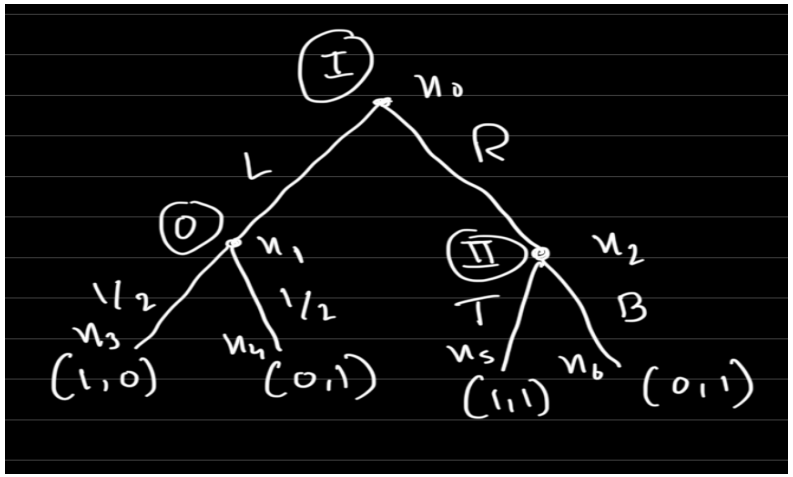
- In a game without chance moves, a strategy vector determines a unique play of the game, and hence unique utilities.
- However, when a game includes chance moves, a strategy vector determines a probability distribution over different plays of the game, and hence a probability distribution over different utility vectors.

Utilities at a strategy profile

- Consider the following game:

$u_1(L,T) = \frac{1}{2}$

$u_2(L,T) = \frac{1}{2}$



	T	B
L	(1/2, 1/2)	(1/2, 1/2)
R	(1,1)	(0,1)

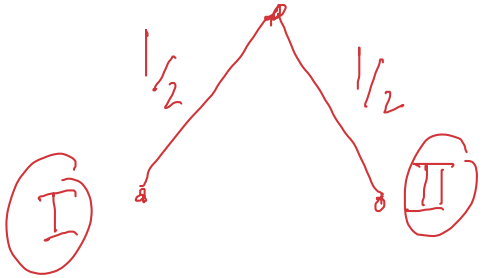
Figure: Figure 2

- The strategy profile (L, T) determines the probability distribution over the end nodes: both x_3 and x_4 are reached with $\frac{1}{2}$ probabilities.

Utilities at a strategy profile

- We define the utility at (L, T) as the expected utility according to the probability distribution induced by the strategy profile.

von Neumann's Theorem?

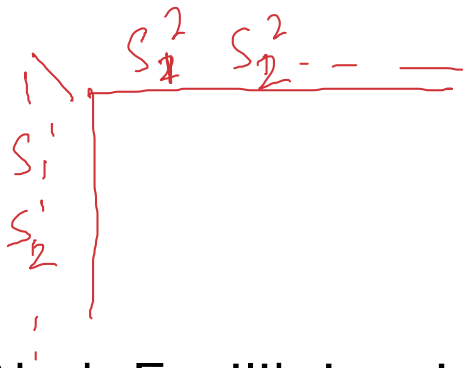


- Theorem 1 does not hold in games with chance moves.
- Think about the Chomp game, where, before the game, a toss happens, and accordingly, the first mover is chosen.

Card games

- Most card games, such as Poker or bridge, cannot be represented in this form, as before the beginning of the game, cards are distributed randomly, and players do not know what cards the other players are holding.

Nash Equilibrium (in pure strategies)



- Keeping in mind the same intuition of Nash Equilibrium in the strategic form game, we can define NE in extensive form games as well.
- A strategy profile $s = (s_1, \ldots, s_n)$ is an NE in the extensive form game if for all $i \in N$,

$$u_i(s) \geq u_i(s_i', s_{-i}) \text{ for all } s_i' \in S_i.$$

Example

- Consider the version of the battle of the sexes game where the man makes the first choice and the woman sees that choice before making her move.

	(F,F)	(F,B)	(B,F)	(B,B)
F				
B				

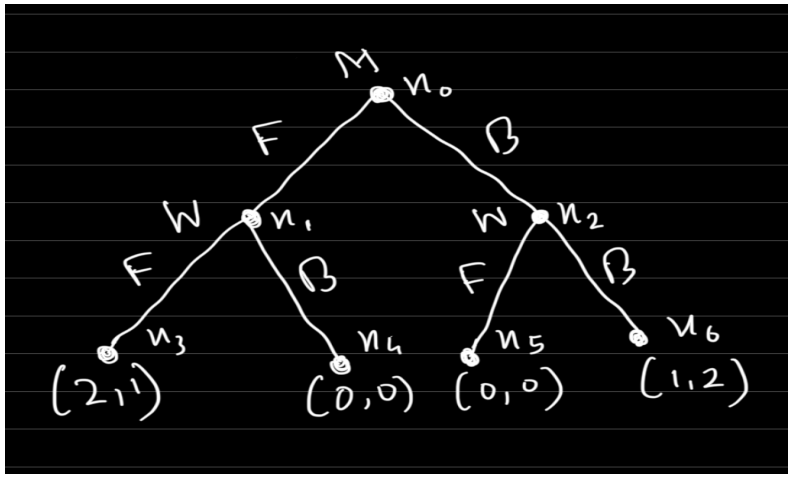


Figure: Figure 3

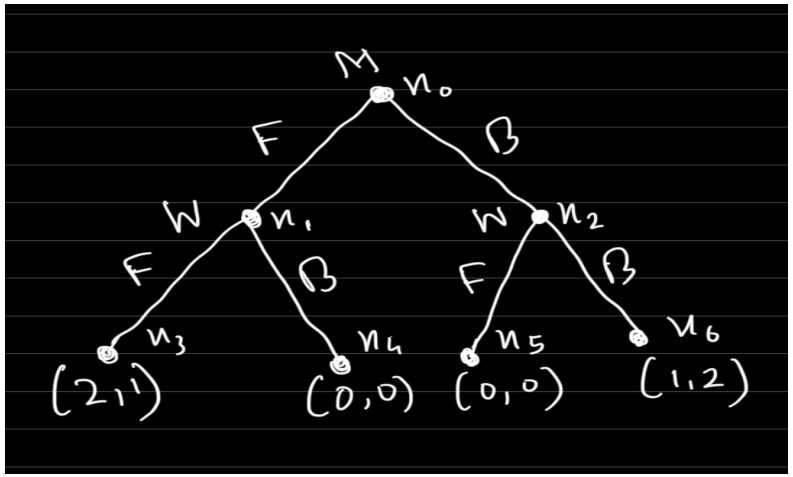
- This game has three NE, namely, $(F,(F,F))$, $(F,(F,B))$ and $(B,(B,B))$ (verify!).

Subgame Perfect Equilibrium (in pure strategies)

- A strategy vector $s = (s_1, \ldots, s_n)$ in an extensive form game Γ is a subgame perfect equilibrium if for every subgame of Γ, the restriction of the strategy vector s to that subgame is a Nash equilibrium in that subgame.
- Since every game is a subgame of its own, every subgame perfect equilibrium is also an NE of the whole game.
- The concept of subgame perfect equilibrium is therefore a refinement of the concept of Nash equilibrium.

Example

- Consider the previous game



$(F,(F,B))$

Figure: Figure 4

- Recall that this game has three NE, namely, $(F,(F,F))$, $(F,(F,B))$ and $(B,(B,B))$.
- Out of the three, only $(F,(F,B))$ is a subgame perfect equilibrium.

Backward induction principle

- We can analyze the extensive form games with chance moves using the principle of backward induction.
- This means that we start with the nodes preceding the end nodes, and turn them into end nodes with payoffs resulting from choosing the optimal action(s) (giving maximum utility to the player taking action at that node). If there are two or more such nodes, the player may arbitrarily choose one of them.
- If one of these nodes is a node of a chance move, then the utility at that node is the expectation of the utilities at the leaves reached by the chance move.

Backward induction principle

- In this way, we obtain a reduced game, and continue the process until we reach the root.
- In some cases, this process may yield multiple strategy profiles.
- Such a strategy profile is called a backward induction equilibrium.

Example

- Consider the following game:

$((c,a),(r,l',R))$

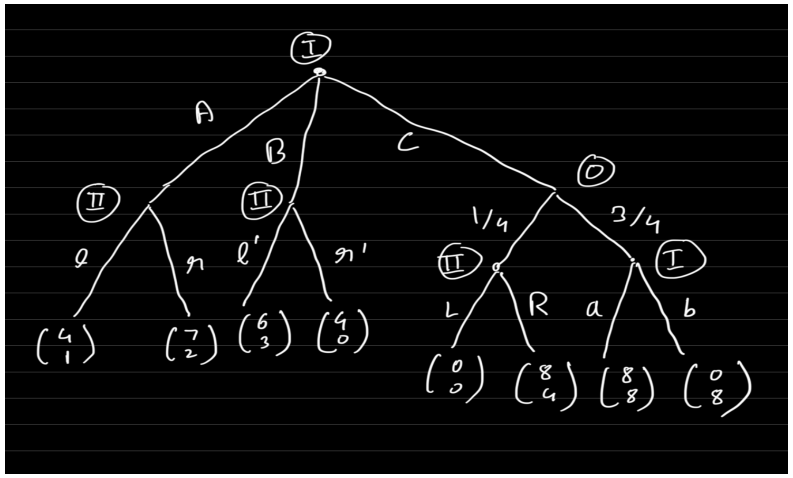


Figure: Figure 5

- If we apply backward induction, Player II will choose the strategy (r, l', R) and Player 1 (at the right-most node) will select a.

Example

- Thus, we have the following reduced game:

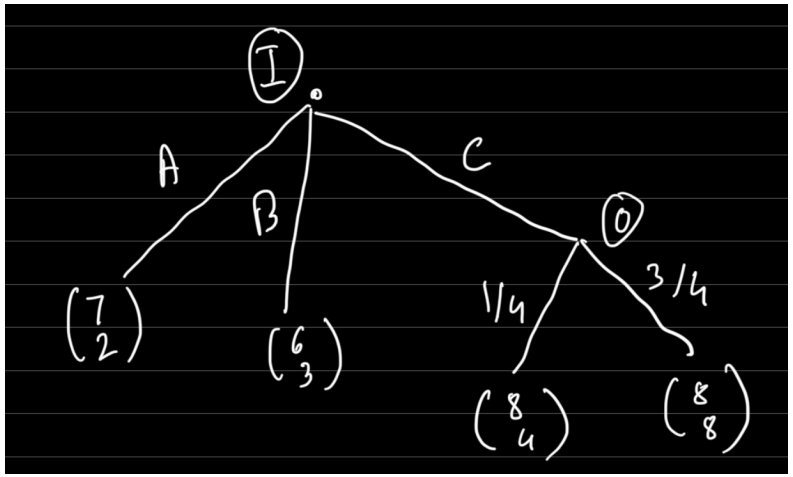


Figure: Figure 6

- In the reduced game, Player I will select the action(s) that will give him maximum utility. Playing A will give 7, playing B will give 6, and if he plays C, the expected utility will be $8\frac{1}{4} + 8\frac{3}{4} = 8$. Hence, he will choose action C.

Example

- Thus, the unique backward induction equilibrium is $((C, a), (r, l', R))$.